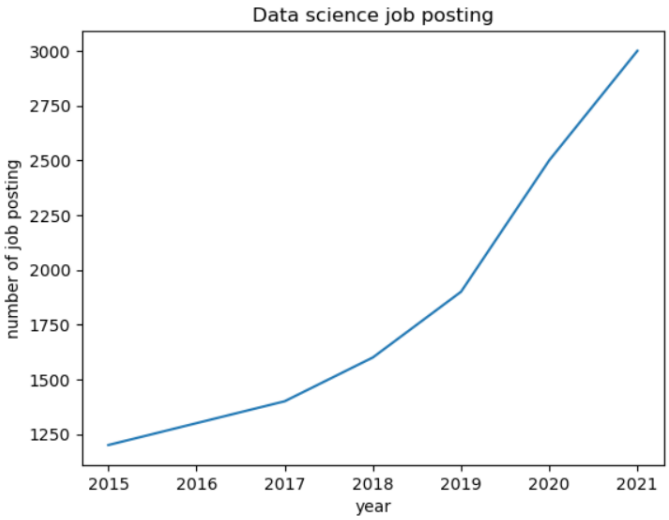
FUNDAMENTAL OF DATASCIENCE

1.a. Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib for visualization.

Program:

```
import matplotlib.pyplot as plt
x=[2015,2016,2017,2018,2019,2020,2021]
y=[1200,1300,1400,1600,1900,2500,3000]
plt.plot(x,y)
plt.title('Data science job posting')
plt.xlabel('year')
plt.ylabel('number of job posting')
plt.show()
```

Output:



Result:

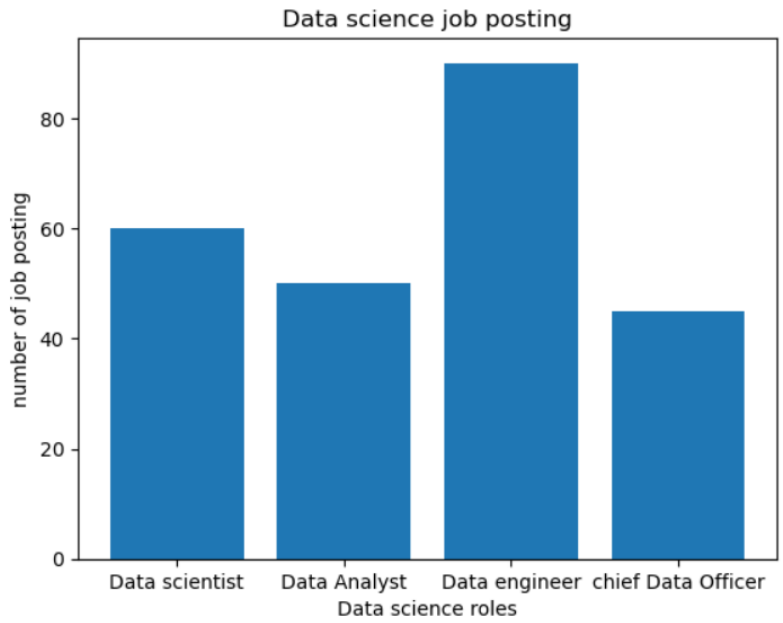
The trend of data science job posting over the decade was analyzed.

1.b.Analyze and visualize the distribution of various data science roles (Data Analyst,Data Engineer, Data Scientist, etc.) from a dataset.

Program:

```
import matplotlib.pyplot as plt
x=['Data scientist','Data Analyst','Data engineer','chief Data Officer']
y=[60,50,90,45]
plt.bar(x,y)
plt.title('Data science job posting')
plt.xlabel('Data science roles')
plt.ylabel('number of job posting')
plt.show()
```

Output:



Result:

The distribution of various data science roles was analysed.

1.c.Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data setsgiven.

Program:

```
import pandas as pd
structured_data = pd.DataFrame({
    'ID': [101, 102, 103],
    'Name': ['John', 'David', 'Emma'],
    'Age': [22, 28, 24]
})
print("Structured Data:\n", structured_data)
unstructured_data = "The weather today is sunny. It may include text, images, audio, or video."
print("\nUnstructured Data:\n", unstructured_data)
semi_structured_data = {
    'ID': 101,
    'Name': 'John',
    'Attributes': {
        'Height': 170,
        'Weight': 65
    }
}
print("\nSemi-structured Data:\n", semi_structured_data)
```

Output:

```
Structured Data:
     ID   Name  Age
0  101   John   22
1  102  David   28
2  103   Emma   24

Unstructured Data:
 The weather today is sunny. It may include text, images, audio, or video.

Semi-structured Data:
 {'ID': 101, 'Name': 'John', 'Attributes': {'Height': 170, 'Weight': 65}}
```

Result:

An experiment to differentiate Structured , Unstructured and semi-structured data was conducted.

1.d.Conduct an experiment to encrypt and decrypt given sensitive data using the cryptography library fernet.

Program:

```
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Software engineering")
print(token)
f.decrypt(token)
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Computer Science Engineering."
cipher_text = cipher_suite.encrypt(plain_text)
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
```

print("Decrypted Data:", decrypted_text)

Output:

```
b'gAAAAABqex_MoGb5m1umJvATBsWMDu9E6CDnefQWokwxoXMu02U_D8XQJvLyKdKgJvU-IpX4UK84NKXXLmkmEhOWvXmkFI5oTS9DbppxjuWqNycakcGDKjk='
Original Data: b'Computer Science Engineering.'
Encrypted Data: b'gAAAAABqex_M0Kxf5sY6cMuxlsocTS8j0s4javPNgHxru-Ar8CcWcjlHAxx_WjxIvKOyh0wmEWnWJcNFBAMX09M0KtGYzEeHRuQyc61EMwx9_eHpJ2IEQQw='
Decrypted Data: b'Computer Science Engineering.'
```

Result:

An Experiment to encrypt and decrypt the data was conducted.